Prodigy Anesthesia Quick Review

Controlled Hypothermia

What is controlled hypothermia and why is it used?	Controlled hypothermia is a reduction in core body temperature that is induced to decrease the cerebral metabolic rate, which provides protection to the brain during periods of ischemia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1007.*
What is the difference between mild and profound hypothermia?	Mild hypothermia is considered to be a reduction in core body temperature to 33-35 degrees Celsius. Profound hypothermia is considered to be as low or even lower than 20 degrees Celsius. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1007.*
Is mild hypothermia of benefit in surgical patients?	Although core temperatures as low as 20 degrees Celsius have been used to successfully provide neurologic protection during cases of total circulatory arrest, even mild hypothermia (33-35 degrees Celsius) demonstrates some potential for neurologic protection by maintaining ATP levels at normal even when oxygen levels are at half normal. Perhaps more importantly, there is a great deal of research that demonstrates that hyperthermia is extremely deleterious for patients who have suffered a cerebral infarct, subarachnoid hemorrhage, or cardiac arrest. Because of this information, many clinicians prefer to induce mild hypothermia simply to avoid inadvertant hyperthermia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1007-1008.*
What are the disadvantages of using profound hypothermia?	Cardiopulmonary bypass is needed during the cooling and warming periods and hypothermia-induced coagulopathy can present concerns for post-operative bleeding. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1007.*
What is the typical application of profound hypothermia?	Some cardiac and cerebral procedures such as aneurysm repair can be performed under complete circulatory arrest using profound hypothermia. When the body temperature is less than 20 degrees Celsius, circulatory arrest can be tolerated for up to about 30 minutes. Profound hypothermia provides greater protection against cerebral ischemia than oxygen administration or anesthetic agents. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1007.*

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